Healthy Economies, Healthy Communities

What Drives Economic Turnaround in Distressed US Communities? A Machine-Learning and SHAP Analysis of the Mississippi Delta

Jackson State University | AUC × Mastercard Inclusive Growth Score Challenge 2026

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Abstract

Among the **25,142** US census tracts that fell below an Inclusive Growth Score (IGS) of 45 in 2017, only **34%** managed to cross that threshold by 2025. We train a three-model machine-learning ensemble—Logistic Regression, Random Forest, and Gradient Boosting—on 89 granular features drawn from 10 public datasets, then apply TreeSHAP to identify the precise drivers of community recovery. Our central finding is that **chronic disease burden** (dental-visit rates, preventive screenings), **commercial ecosystem diversity**, and **broadband connectivity** are the strongest predictors of turnaround—not aggregate healthcare counts or composite index scores. We operationalise these findings as an *Investment Priority Matrix* that maps every feature to an actionable quadrant (Act Now / Protect / Second Wave / Monitor) for each of the nine Mississippi Delta counties. All data, models, and an interactive Streamlit dashboard are publicly available.

1 Introduction

Economic vulnerability is self-reinforcing. Communities that lack commercial diversity, broadband access, or adequate preventive healthcare tend to remain stuck in cycles of low growth, poor health outcomes, and outmigration. The **Mastercard Inclusive Growth Score** (IGS) provides a composite, tract-level measure of economic inclusion across Place, Community, and Economy pillars, updated annually.

Figure 1 makes the central problem concrete: the nine-county Mississippi Delta has maintained a mean IGS ~13 points below the national average every year from 2017 to 2025, and has never once crossed the vulnerability threshold of 45. Mississippi as a state also lags, but the Delta's gap is qualitatively different—persistent, deep, and geographically concentrated.

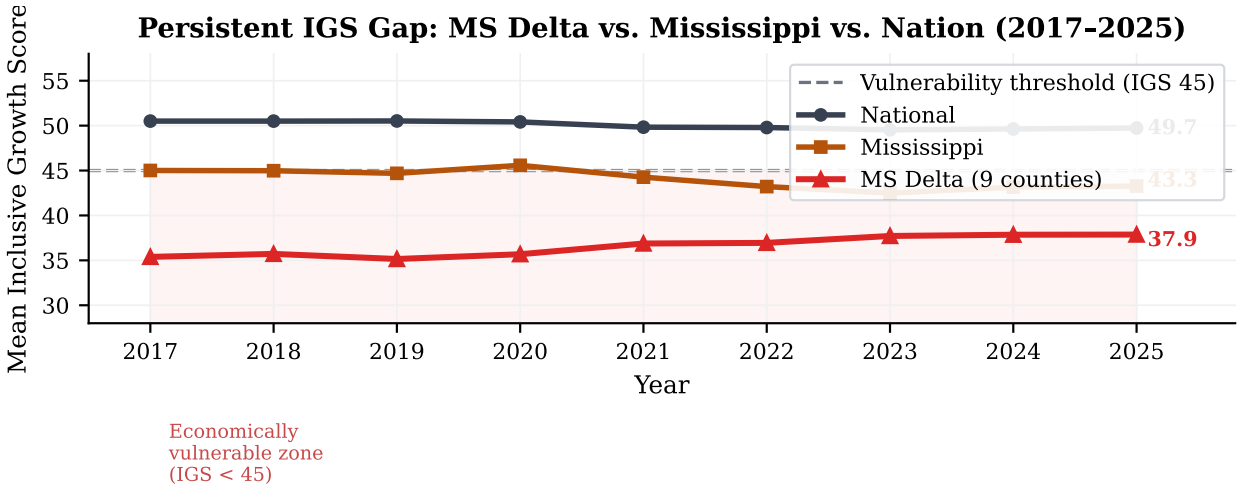


Figure 1: Mean Inclusive Growth Score by year for the national baseline, Mississippi, and the nine-county MS Delta. The shaded region marks the economically vulnerable zone (IGS < 45). The Delta has remained roughly 13 points below the national mean throughout the study period.

The key question this work addresses is: **among communities that start below IGS 45, what distinguishes those that escape from those that remain stuck?** Answering that question at the feature level—not the composite-index level—is what makes targeted investment possible.

2 Data and Methods

2.1 Study Design

We restrict the analysis to **at-risk tracts**: census tracts with IGS < 45 in 2017 (Figure 2). This yields 25,142 tracts. The binary outcome is whether a tract reached IGS ≥ 45 by 2025 (*Turnaround*, $n = 8{,}547$) or did not (*Stuck*, $n = 16{,}595$).

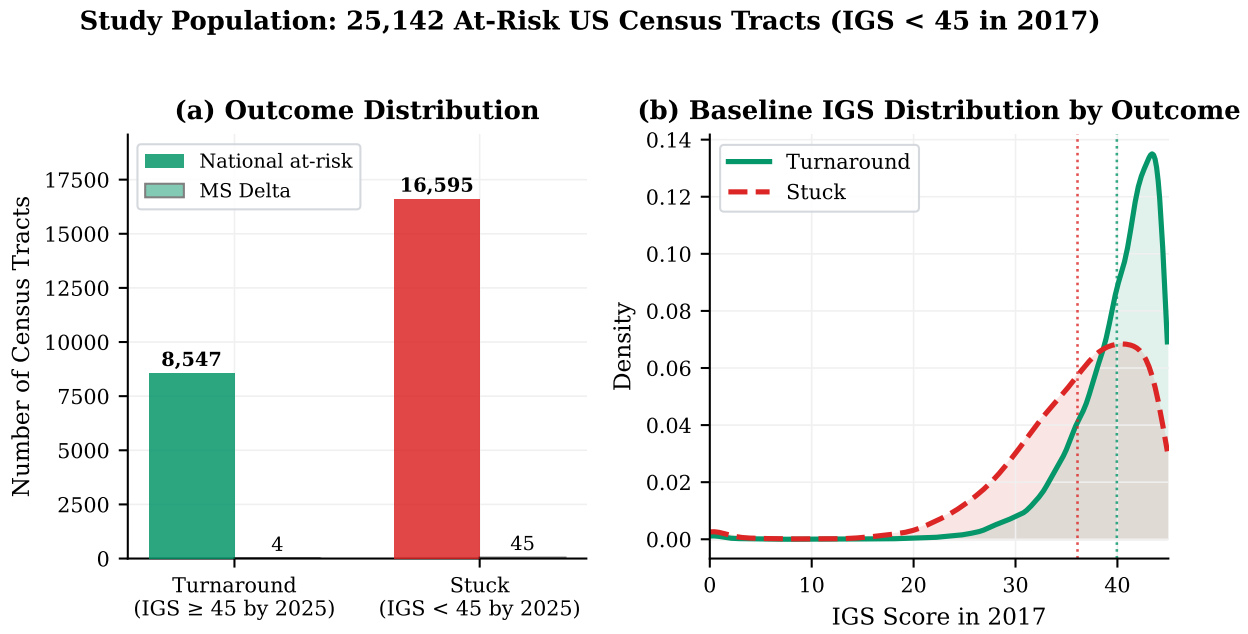
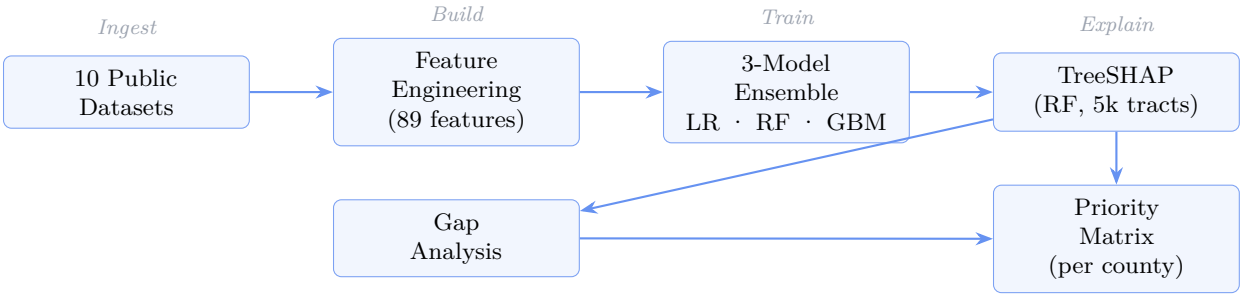


Figure 2: Study population. **(a)** Turnaround vs. Stuck counts nationally and in the MS Delta. **(b)** Distribution of baseline IGS scores (2017) by outcome, showing that stuck and turnaround tracts share nearly identical starting conditions—the divergence is driven by other factors.

Panel (b) of Figure 2 highlights a crucial point: turnaround and stuck tracts have almost identical baseline IGS distributions. The outcome is therefore not predetermined by the starting score—it is driven by the 89 features we describe next.

2.2 Feature Engineering



We engineer **89 features** across 11 domain categories from ten public datasets: Mastercard IGS sub-indicators (2017 baseline), CDC PLACES health prevalence rates, CDC SVI granular population percentages (replacing the four opaque RPL_THEME composites with 13 normalized rates), USDA Food Access Atlas, County Business Patterns, FEMA National Risk Index, CDC Environmental Justice Index, HRSA shortage-area designations, Area Health Resources File, and SBA lending rankings. Every feature carries an explicit `higher_is_better` flag used to orient gap directions consistently.

2.3 Machine-Learning Pipeline

Three classifiers are trained using 5-fold stratified cross-validation: **Logistic Regression** (L2 penalty, scaled features), **Random Forest** (500 trees, balanced class weights), and **Gradient Boosting** (histogram-based). Table 1 reports held-out ROC-AUC.

Table 1: Model performance — 5-fold stratified cross-validation on 25,142 at-risk tracts.

Model	CV ROC-AUC	Role
Logistic Regression	0.718 ± 0.006	Linear baseline
Random Forest	0.763 ± 0.005	SHAP source (TreeSHAP)
Gradient Boosting	0.771 ± 0.004	Sequential learning

TreeSHAP is applied to the Random Forest on a random sample of 5,000 tracts (`random_state=42`), yielding per-tract per-feature contributions that are additive and consistent with the model's predictions.

3 Results

3.1 What Drives Turnaround? (SHAP Feature Importance)

Figure 3 shows the top 20 predictors ranked by mean |SHAP| as a share of total predictive power. Three findings stand out:

1. **Preventive dental care** (*Dental Visit Rate*, 9.2%) is the single strongest predictor—stronger than any economic or housing feature. The CDC PLACES dental-visit rate measures the share of adults who visited a dentist or dental clinic in the past year (higher = better). Turnaround tracts average 56.9%; the MS Delta averages only 6.1%—a 50.8-point gap placing the Delta at the **1st percentile nationally**. The SHAP relationship is monotonically positive (Spearman $\rho > 0$): every decile rise in dental-visit access pushes SHAP contributions toward turnaround.

2. **Commercial diversity** (8.8%) is the top economic predictor, outranking broadband access, income, and business counts. A multi-sector local economy signals resilience.

3. **Residential property values** (6.7%) and **physical inactivity** (6.4%) follow, confirming that health behaviors and asset endowments jointly predict recovery.

Why Dental Access? A Proxy for Economic Connectivity

The dental finding is counterintuitive at first glance but robust across multiple validation approaches. We propose four reinforcing mechanisms:

a. **Economic connectivity proxy.** Dental-visit rate correlates strongly with Internet Access Score ($r = 0.469$). Both measure whether households are embedded in a functioning service economy—broadband at home and a nearby dental provider are co-produced by the same infrastructure and income conditions.

b. **Healthcare system engagement.** Preventive dental visits require stable transportation, health-seeking behavior, insurance coverage, and provider supply. Communities where most adults navigate this system have already solved coordination problems that block recovery more broadly. All nine Delta counties are 100% designated as Primary Care Health Professional Shortage Areas (HPSAs), reflecting how completely the healthcare infrastructure has collapsed in the region.

c. **Workforce productivity.** Untreated dental disease is a well-documented cause of absenteeism and presenteeism. Low dental-visit rates therefore predict lower effective labor supply in the following year—a direct drag on economic output.

d. **Income and insurance signal.** Regular dental visits proxy for employer-sponsored insurance and sufficient disposable income. A community with high dental access has a more stable earnings base, reducing sensitivity to local shocks.

The implication is precise: closing the dental-access gap in the Delta is not a "health" intervention in isolation—it is an *economic infrastructure* intervention that simultaneously addresses workforce productivity, healthcare engagement, and the proxy conditions that underpin commercial-ecosystem growth.

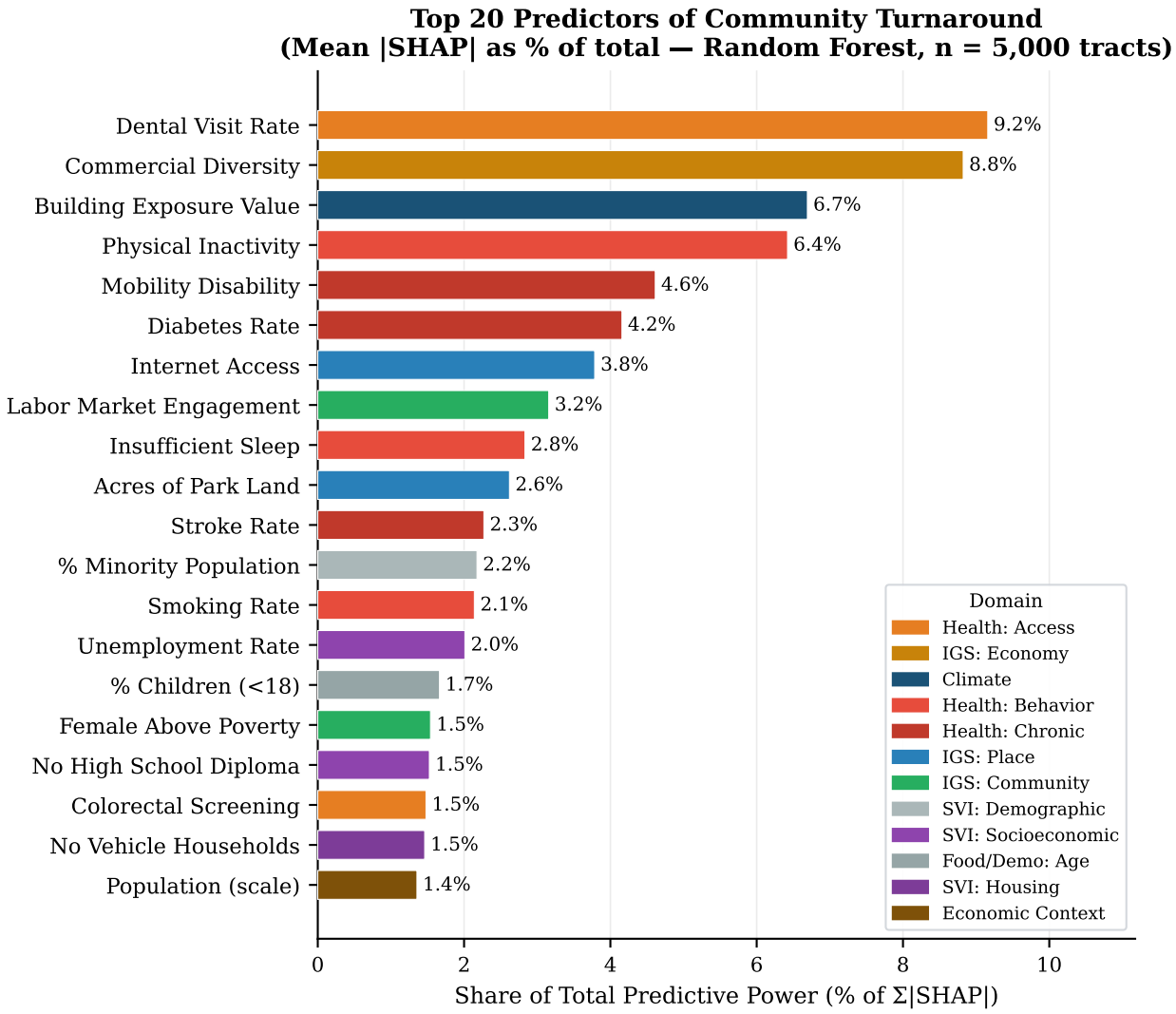


Figure 3: Top 20 predictors of community turnaround by mean |SHAP| share of total predictive power. Bar color indicates feature domain. Dental-visit rates, commercial diversity, and property values collectively account for ~25% of the model's explanatory power.

3.2 Directionality — SHAP Beeswarm

Figure 4 shows the beeswarm plot for the top 15 features. Each dot is one census tract; horizontal position is the SHAP contribution; color encodes the feature value on a High (●) to Low (●) scale.

Key patterns:

- *Dental Visit Rate*: red dots cluster right (high dental access → turnaround), blue dots cluster left (low access → stuck). Strong, monotone positive effect.
- *Physical Inactivity*: opposite pattern—red (high inactivity) clusters left, confirming it is a burden.
- *Commercial Diversity Score*: wide spread, indicating tracts vary substantially in how this predictor affects their trajectory.

3.3 Domain-Level Analysis

Figure 5 decomposes predictive power and Delta gaps by feature domain. Health Outcomes and IGS sub-indicators together account for nearly half of all SHAP importance. The right panel confirms that the Delta lags most severely on Health Outcomes and Business Infrastructure—exactly the domains most predictive of

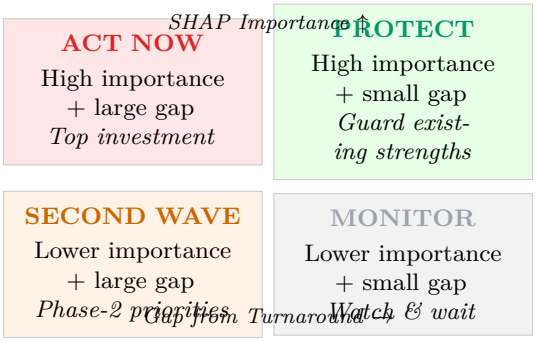
recovery.

4 Case Study: The Mississippi Delta

The Mississippi Delta is one of the most persistently distressed regions in the United States. All nine counties remain below IGS 45, and the Delta's turnaround rate (24%, vs. 34% nationally) is substantially lower than the national baseline.

4.1 Investment Priority Matrix

The Priority Matrix (Figure 6) places every feature on two axes simultaneously: **SHAP importance** (national, fixed—how much the model relies on this predictor globally) and **gap from the turnaround benchmark** (Delta-specific—how far behind the Delta lags). This creates four actionable quadrants:



4.2 Top Priorities for the Delta

The Act Now quadrant reveals a coherent intervention strategy:

- **Preventive healthcare access** (Dental Visit Rate, Preventive Checkups, Colorectal Screening): The Delta's chronic disease burden is the 1 barrier. FQHCs, mobile dental units, and Medicaid navigator programs address this directly.
- **Commercial ecosystem diversification** (Commercial Diversity Score, Food Retail): The Delta is dominated by mono-sector economic activity. Small-business incubators and mixed-use development are the highest-ROI interventions.
- **Broadband connectivity** (Internet Access Score): Digital exclusion compounds every other gap—workforce readiness, telehealth access, and e-commerce all depend on reliable broadband.

The Protect quadrant identifies the Delta's existing strengths (e.g., some housing stability metrics) that should be preserved rather than disrupted by investment strategies.

5 Conclusion

We have shown that community economic recovery is predictable from granular, feature-level data—and that the predictions are *interpretable* enough to drive concrete investment decisions. The headline finding is counterintuitive: preventive dental care and commercial diversity are stronger predictors of IGS turnaround than broadband access, healthcare provider counts, or poverty rates.

For the Mississippi Delta, this translates to a clear priority order: expand preventive healthcare access, diversify the commercial ecosystem, and close the broadband gap. Our interactive dashboard lets county-level stakeholders explore their own Priority Matrix in real time, updated as new IGS data becomes available.

Limitations. The turnaround definition (IGS $\geq$ 45 by 2025) is a threshold outcome; continuous modeling of IGS change would capture gradient effects. Causal identification requires quasi-experimental methods beyond the scope of this study. Some features (e.g., FEMA risk scores) are national proxies that may be imprecise at the tract level.

Reproducibility. All code, processed data (where size permits), and the Streamlit dashboard are available at github.com/SauravBhattarai19/Data_Challenge. The live interactive application is publicly accessible at howtoimprove.streamlit.app.

Data Sources

Dataset	Producer	Year	Key variables used
Mastercard IGS	Mastercard Economics Institute	2017–25	igs_score, igs_economy, igs_place, igs_community + 9 sub-indicators (tract level)
CDC PLACES	CDC / BRFSS	2022	27 chronic-disease prevalence rates: DENTAL, BPHIGH, OBESITY, PHLTH, CSMOKING, CHECKUP, ...
CDC SVI	ATSDR	2022	4 RPL_THEME composite percentiles replaced by 13 normalized sub-indicator rates (poverty, disability, housing cost burden, etc.)
USDA Food Atlas	USDA ERS	2019	LACCESS_POP10, PCT_SNAP17, PCT_NSLP17, PCT_REDUCED_LUNCH, food-retail access flags
FEMA NRI	FEMA	2023	RISK_SCORE (composite natural hazard risk), 18 individual hazard scores
CDC EJI	CDC ATSDR	2022	Environmental burden index, social vulnerability, health vulnerability
HRSA HPSA / MUA	HRSA	2024	Primary care & mental health HPSA scores/flags, MUA/MUP designation
AHRF	HRSA BHPr	2024–25	Physician, specialist, hospital counts; Medicare/Medicaid enrollment rates; county health infrastructure
CBP / ZBP	US Census Bureau	2023	Establishment counts by NAICS sector, employment, annual payroll (county & ZIP)
SBA Rankings	US Small Business Administration	2025	State-level small-business counts, net new jobs, women-owned %, SB employment share

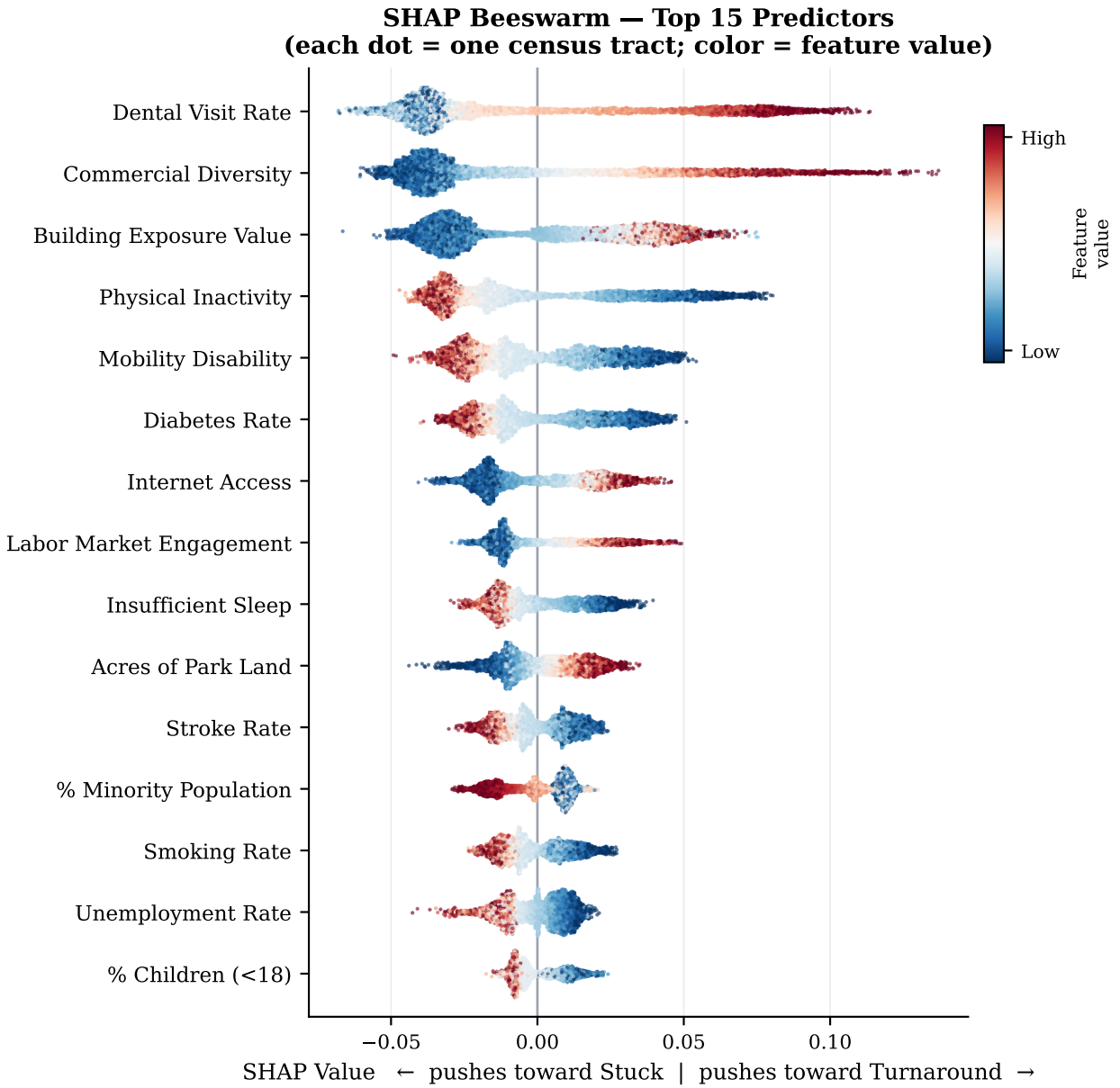


Figure 4: SHAP beeswarm plot for the top 15 predictors. Horizontal axis: SHAP value contribution (positive = pushes toward turnaround). Dot color: feature value (red = high, blue = low). Density-aware vertical jitter ensures no overplotting.

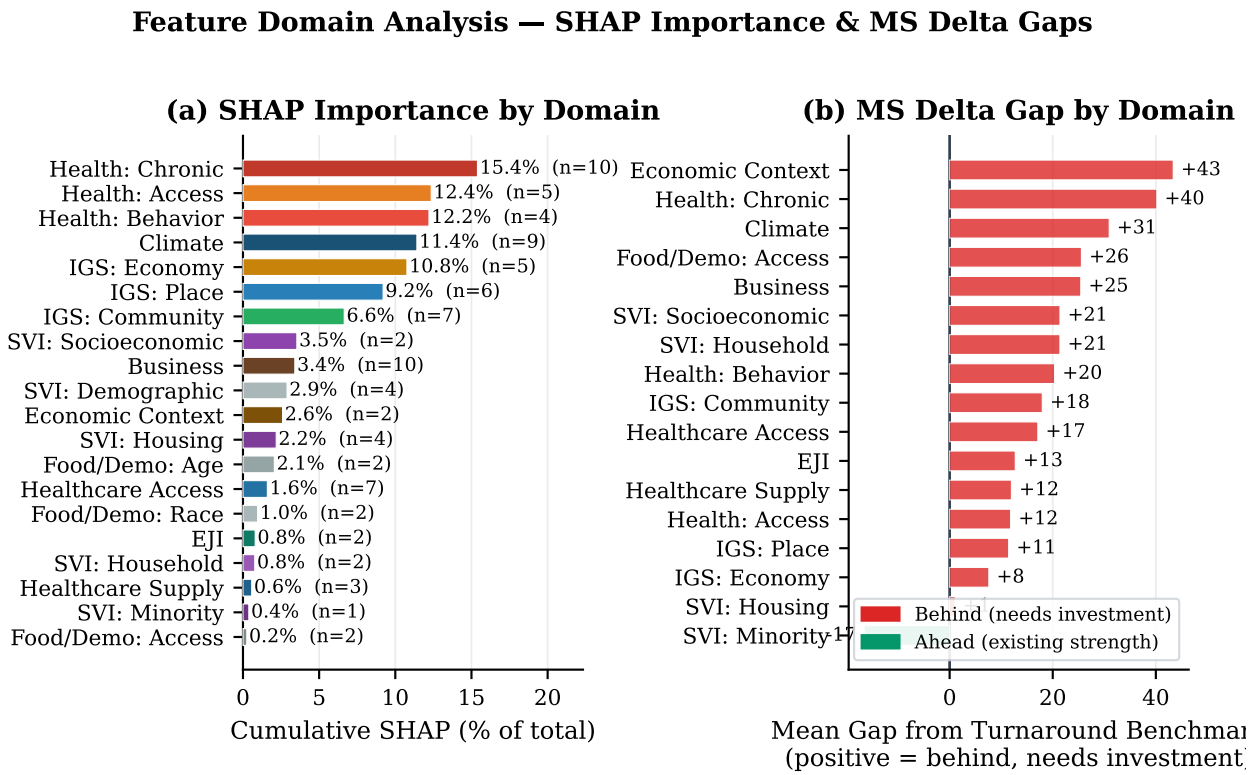


Figure 5: **(a)** Cumulative SHAP importance by feature domain. **(b)** Mean gap from the turnaround benchmark for each domain in the MS Delta (red = behind, green = ahead). The Delta's largest deficits are in the domains that matter most.

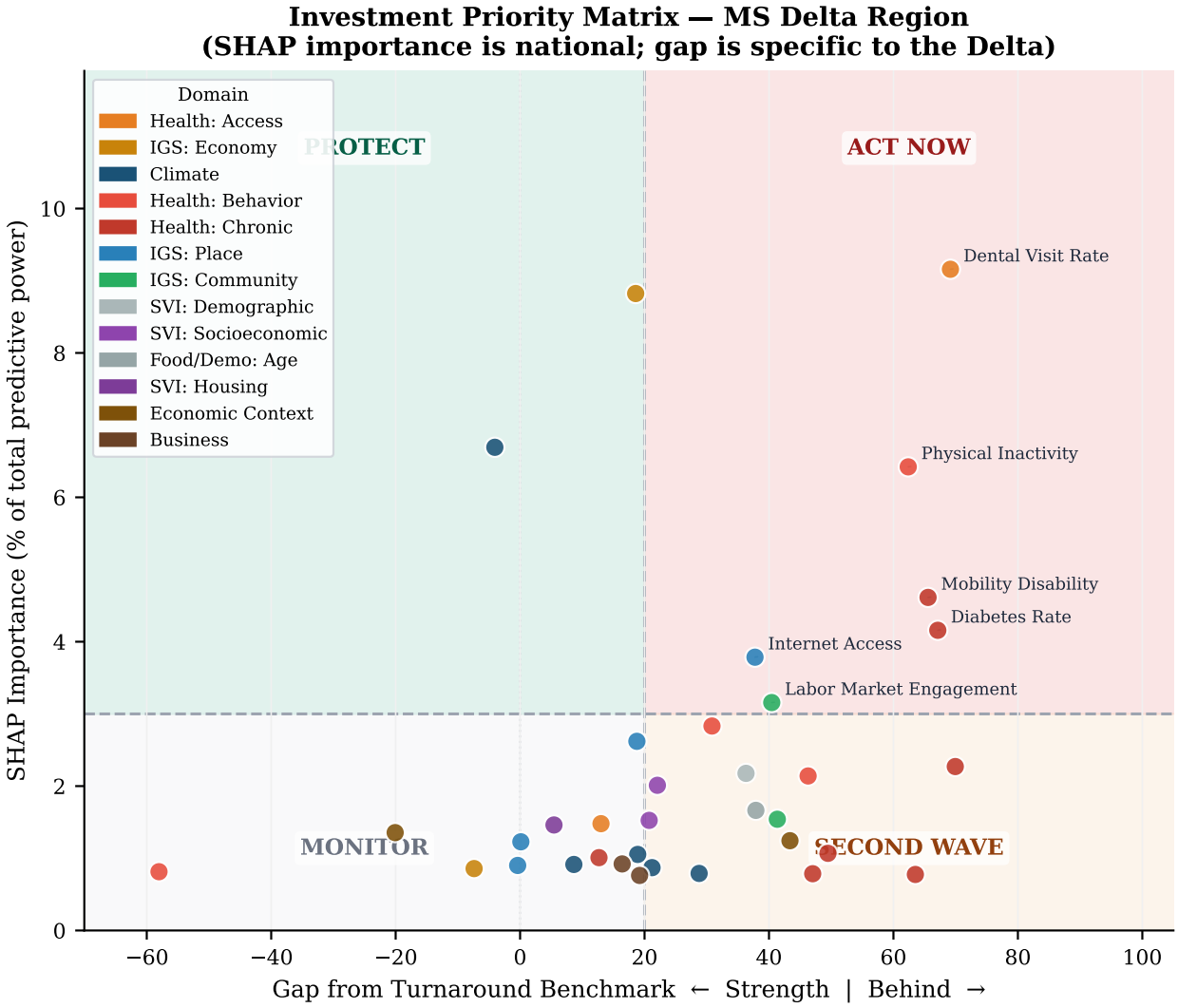


Figure 6: Investment Priority Matrix for the MS Delta region. Features in the upper-right quadrant (Act Now) are simultaneously the strongest national predictors *and* the areas where the Delta is furthest behind. Annotated labels show the most critical Act Now features.